

Timothée Ly

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EDUCATION

Agrégé de Mathématiques

France | 2024

EDUCATION NATIONALE

The most competitive and prestigious examination for civil service in the French public education system.

Diplôme d'Ingénieur (Master equivalent)

Palaiseau, France | Sept 2020 - Sep 2024

TELECOM PARIS | GPA: 4.0/4.0

MEMBER OF INSTITUT POLYTECHNIQUE DE PARIS (IP PARIS) – 48th QS RANKING, 14th WORLDWIDE IN MATH 2023

Coursework: Stochastic Calculus, Continuous Time Markov Chains And Martingales, Statistics, Analysis (Fourier & Functional), Optimization, Machine Learning, Representation of Signals, Time Series, Speech and Audio Processing

Double Degree

Stockholm, Sweden | Aug 2022 - Sep 2024

KTH ROYAL INSTITUTE OF TECHNOLOGY

Coursework: Data Mining, Data Intensive Computing, Advanced Machine Learning, Artificial Neural Networks and Deep Architectures, Natural Language Processing, Information Retrieval Systems

Classes Préparatoires

Toulouse, France | Sept 2018 - Jun 2020

LYCÉE PIERRE DE FERMAT | GPA: 4.0/4.0

Two intensive years of mathematics, physics and computer science to prepare for the national competitive exams of the *grandes écoles*. Fermat is one of the most prestigious *classes préparatoires* in France.

WORK EXPERIENCE

RESEARCH ON DIFFERENTIAL PRIVACY | INTERNSHIP

LAAS-CNRS, France | Sept 2023 - Feb 2024

- Determined a tighter bound on Gini impurity sensibility and achieved higher accuracy with Smooth Sensibility on Greedy Rule Lists. [↗](#)
- Implemented a Differentially Private version of Corels with Laplace Mechanism. [↗](#)
- Paper available on Hal (submitted to NIPS24) [↗](#)

PROJECTS

NEWS RECOMMENDATION SYSTEM [↗](#)

ELASTICSEARCH, BOOLEAN QUERY, TF-IDF, COSINE SIMILARITY

A news recommendation system built on top of Elastic Search which offers boolean query retrieval and recommendations processed from the user's preferences.

NETWORK INTRUSION DETECTION [↗](#) [↗](#)

DECISION TREES, NEURAL NETWORKS, NSL-KDD, IDS2017

Two-week project aimed at improving existings Neural Network systems trained to detect network intrusion.

VOLUNTEERING

FORUM ASSOCIATION | IT SECTION

Palaiseau, France | 2021 - 2022

The Telecom Paris Forum is the association responsible for the organisation of the fair where companies come to the school to meet students. As a member of the IT section, I was in charge of maintaining the website and the database.

MATHEMATICS AND PHYSICS TUTORING | TUTOR FOR HIGHSCHOOLERS

Remote | 2021 - 2024

SKILLS

Languages: Java, C/C++, Python, SQL, Notions of JavaScript, HTML/CSS

Technology: Git, MySQL, Elasticsearch, Spark, $\LaTeX$

Languages: French (Native speaker), English (TOEFL iBT: 110/120), Spanish (B1), Chinese(B1)